Soheil Sayah Varg

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Education

Sharif University of Technology **Tehran, Iran**
B.Sc. in Computer Engineering *Sep 2023 – Jul 2027*

- **Cumulative GPA: 19.38 / 20.0**, ranked **15th** of 183 students. Departmental average for this cohort: **17.24**; university-wide: 15.85.
- Graduate coursework taken as an undergraduate: **Stochastic Processes** (20.0), **Reinforcement Learning**.
- Awarded 7 credits in physics on admission under the National Olympiad medalist provision.

AE High School **Tehran, Iran**
Diploma, Mathematics & Physics *Sep 2020 – Jun 2023*

- GPA: 19.25 / 20.0

Honors and Awards

2023: **Gold Medal**, Iranian National Physics Olympiad — ranked **7th** nationwide among more than 10,000 participants, and **2nd** nationwide in the experimental (laboratory) examination.

2023: Admitted to Sharif University of Technology through the olympiad medalist track (no entrance exam).

Research Interests

Primary: **Inference and approximation under uncertainty** — when the object of interest cannot be computed exactly, what licenses trusting the approximation. Concretely: exact and approximate sampling for language models, and estimation for offline decision-making under partial observability.

Foundations: Statistical learning theory, reinforcement learning theory, and convex optimization — the tools these questions get asked in, and the source of the gap between what an estimator provably does and what it does at finite sample size.

Research Experience

Data Science & Machine Learning Laboratory (DML), Sharif University of Technology
Exact Controlled Generation from Tilted Autoregressive Targets *Dec 2025 – Present*

- Supervised by **Dr. Ali Rostami**.
- Sampling scheme that draws from a reward-tilted language-model distribution *exactly in distribution* rather than approximately, via a particle-Gibbs ladder over a batched proposal chain — so that steering a model's output does not silently change which distribution is being sampled from.
- Replaced the design's assumed theoretical scaling rate with a measured one across sequence lengths spanning a 32× range, which moved the method's usable operating regime and re-anchored the design on measured rather than extrapolated points.
- **First-author manuscript in preparation.**

Graduate Deep RL project, team of 3, Sharif University of Technology
Model-Based Reinforcement Learning in Confounded POMDPs *Feb 2026 – Present*

- Built the empirical implementation of Hong, Qi & Xu (ICML 2024), a theory-only paper shipping no code or experiments: dual bridge-function identification (reward and dynamics) from pre-decision negative-control proxies, verified against an analytic oracle to machine precision before any result was trusted.
- Found the paper's pessimism step *diverges* numerically — then found our own implementation had dropped a bridge-class constraint the theorem carries, which accounts for part of the magnitude. The divergence survives restoring it, and our signal-subspace repair helps only mid-grid, so it stands as a disclosed heuristic rather than a certified bound.
- Ran a 5-seed sweep at full benchmark scale and found the **confounding-blind baseline wins in all 15 cells** — estimation variance dominates the bias the method removes. Extending to continuous state and action through an RKHS backend with a closed-form linear-Gaussian oracle reverses the verdict.
- Follow-on theory: a **schedule-free lower bound** on the pessimistic width in which the regularizer cancels, so no ridge schedule escapes it, with two switches checkable before fitting — an inadequate negative control leaks the truth, confounding leaks the value gradient.
- *Role in the team of three:* I read the two anchor papers and drove the implementation; the remaining literature and estimator correctness review were shared.

Selected Projects

Information Retrieval: **Modern Information Retrieval System** — Python, PyTorch, FAISS repo

- Three-stage retrieval pipeline: SPLADE learned-sparse and dense retrieval fused by reciprocal rank fusion, then cross-encoder reranking; QLoRA fine-tuning for the generation stage. Evaluated with NDCG and MAP against a documented weak-supervision protocol.

Applied Probability: **Market Predictability Study** — Python, statsmodels repo

- Independent re-examination, carried out after the course had ended, of my own profitable trading backtest — variance-ratio, Hurst, Ornstein-Uhlenbeck MLE, Newey-West with heteroskedasticity-consistent errors. Showed the claimed 5-tick lead signal was a multiple-comparisons artifact (scanning 21 lags of pure noise clears a single-test threshold 69.3% of the time) and that random entry reproduces the strategy's Sharpe ratio. The effect that survives is parameter staleness under drift: trailing re-estimation halves maximum drawdown.

Optimization: **Convex Optimization Algorithms** — Python, NumPy, CVXPY repo

- First-order methods implemented from their update rules, with a solver used only as an oracle to disagree with. Chambolle-Pock primal-dual for TV denoising (adjoint identity verified to 1e-8, step sizes taken from $\tau\sigma\|K\|^2 < 1$ rather than tuned) and ADMM for TV inpainting, which matches CVXPY to 0.02 dB PSNR while running in 0.2 s.

Deep Learning: **Semantic Segmentation and Deep RL** — PyTorch repo

- U-Net, Attention U-Net and Residual-Attention U-Net trained and compared for segmentation; Soft Actor-Critic implemented from scratch (twin critics, reparameterized policy, Polyak-averaged targets) for continuous control.

Systems: **NeoGit** — a version control system in C repo

- Git-like CLI built on raw POSIX file I/O: staging, commits, branching, merge, tagging, stashing, and pre-commit hooks, with an on-disk object layout of its own design.

Teaching Experience

2024 – 2026: **Teaching Assistant**, Sharif University of Technology — **nine course-semesters across five terms**, including **Head TA** for Signals & Systems (Dr. Manzuri, Spring 2026), coordinating the teaching staff for a 60-student core course.

Stochastic Processes and Engineering Probability & Statistics (Dr. Najafi) • Machine Learning and Probability & Statistics (Dr. Sharifi-Zarchi) • Data Structures & Algorithms (Dr. Abam) • Linear Algebra, twice (Dr. Rabiee & Dr. Ramezani) • Fundamentals of Programming (Dr. Rivadeh).

Aug 2023 – Present: **Co-founder and physics instructor**, **Resonance** (resonanceoly.ir) — an online Physics Olympiad school, started as a small study group, reaching students who have no selective high school near them. Teach the mechanics, electromagnetics and laboratory tracks, and write problem sets and mock exams for the national selection rounds.

2022 – Present: **Physics Olympiad instructor**, theory and **experimental / laboratory** tracks — core olympiad staff at **Allameh Helli 3** (NODET, Iran's national programme for exceptional talents) and instructor at AE High School. Also introductory Python at AE Middle School and lower-secondary physics at Mahdavi School.

Academic Service and Outreach

May 2025: **Juror**, 13$^{\text{th}}$ Persian Young Naturalists' Tournament (PYNT), Tehran — an IYPT-format research tournament run by the Ariaian Young Innovative Minds Institute under the World Federation of Physics Competitions. Judged team defenses of open-ended physics research problems.

2024 – 2025: **Deputy Chairman**, CEI 2024 (Computer Engineering Introduction event) • **Head of Purchase**, WSS 11 (Winter Seminar Series) • Sponsorship Staff, WSS 10.

2023 – 2025: **Head of Purchase & Scientific Staff**, Bugsbuzzy game development contest • Executive Staff, **ICPC** 2023 & 2025 • Rayan AI 2024.

Selected Coursework († graduate-level)

- Stochastic Processes†
- Machine Learning
- Artificial Intelligence
- Convex Optimization I
- Linear Algebra
- Probability & Statistics
- Computer Simulation
- Numerical Computation
- Data Structures & Algorithms
- Signals & Systems

All courses above graded 19.5–20.0 out of 20.0.

Spring 2026: Reinforcement Learning†, Modern Information Retrieval, Game Theory, Design of Algorithms, Database Design (*grades pending*).

Technical Skills

Languages: Python (proficient), C, C++, Java, SQL, MATLAB, LaTeX

ML & Numerics: PyTorch, NumPy, SciPy, scikit-learn, statsmodels, pandas, Matplotlib, Hugging Face Transformers

Optimization & IR: CVXPY, FAISS, Gymnasium

Tools: Git, Linux, Docker, PostgreSQL, shell scripting, Proteus